

CAEM: Confidence-Aware Episodic Memory with Self-Improvement

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Abstract

Large language models produce confident but factually incorrect outputs at rates that span three to ninety-one percent across deployment domains, a failure mode that increased parameter count alone does not close. This thesis proposes Confidence-Aware Episodic Memory with Self-Improvement (CAEM), an architectural intervention that treats hallucination reduction as a system-level problem rather than a post-hoc detection problem. A single query passes through a verified episodic memory store, a confidence-aware three-tier router with an independent safety override, and a nine-signal verifier ensemble. Against the matched zero-shot baseline, the within-trajectory best cycle reaches a pooled exact-match gain of +20.3% and a pooled composite hallucination metric reduction of −29.4% on the training panel. Against the locked external eight-baseline panel, CAEM leads on both pooled axes at the best and final cycles.

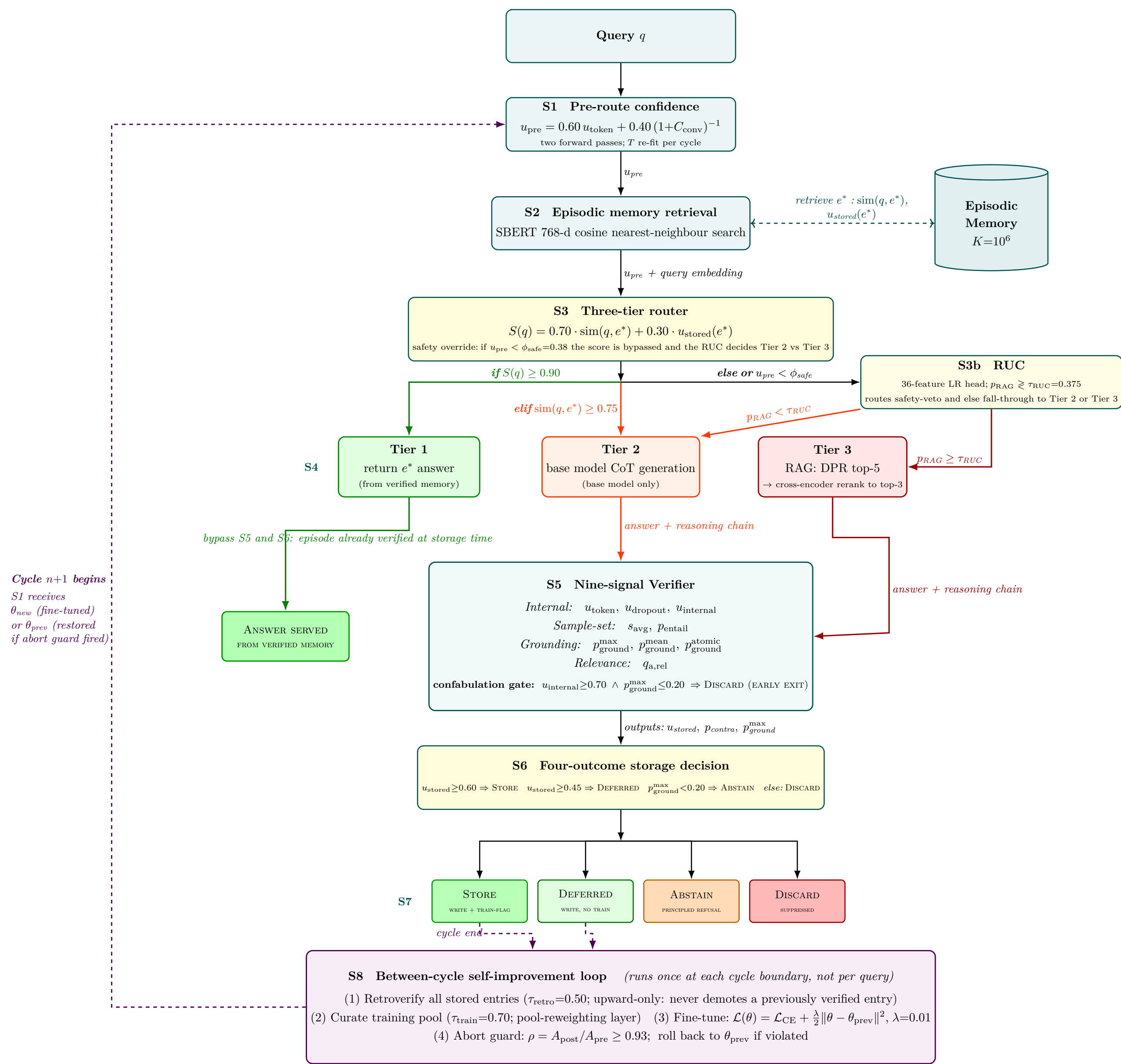
Research Objective

- Verified episodic memory with a four-outcome decision:** design an episodic memory whose admission rule is finer than a single binary threshold.
- Pre-generation confidence, three-tier routing with safety override, and a retrieval-utility classifier:** compute a pre-generation confidence estimate from a single short forward pass on the query that fuses a short-greedy-decode token-probability aggregate with an encoder-layer convergence ratio, calibrate it via per-benchmark temperature scaling on a disjoint fold, and use it independently from the memory similarity score and the stored-answer quality.
- Nine-signal post-generation verifier with a calibrated probability composite:** combine nine complementary signals into a per-answer record across four families: internal calibration of the generator, sample-set agreement across resampled chains, external grounding against retrieved evidence under three aggregation rules, and question-answer relevance.
- Fixed-threshold storage gate with two registered cut-points:** convert the calibrated probability composite into the storage and deferral decisions through a fixed-threshold rule whose two cut-points and one abstention floor are registered before the main run begins and held constant across cycles.
- Constrained self-improvement with a parameter-efficient adapter and a multi-modal retention guard:** curate the fine-tuning pool from stored episodes that exceed a strict quality threshold set above the storage cut-point and pass the curated pool through a benchmark-balancing reweighting layer that prevents single-benchmark dominance.

Methodology

The architecture organises a single query into an eight-stage pipeline traversed once per query and wrapped in a per-cycle update loop. Stage 1 produces a pre-routing confidence from a single short forward pass. Stage 2 encodes the query and probes a similarity-indexed episodic memory for the nearest stored entry. Stage 3 combines the similarity and stored-quality scores into a dispatch score and routes to one of three tiers, with a safety override on low confidence. Stage 3b, a retrieval-utility classifier, refines the dispatch into zero-shot or retrieval-augmented tier per query. Stage 4 executes the chosen tier; Stage 5 passes the answer through a nine-signal post-generation verifier; Stage 6 applies a four-outcome decision (store, defer, abstain, discard); Stage 7 returns the user-facing action. Stage 8, at cycle boundaries, runs retroactive re-verification, deferred reconsideration, memory consolidation, and a regularised fine-tune under a retention guard with asymmetric rollback: weights are rolled back on retention failure, but memory is preserved.

Proposed Model Architecture



The eight-stage pipeline routes each query through pre-routing confidence, memory lookup, three-tier dispatch with the Stage 3b retrieval-utility classifier ($\tau_{RUC} = 0.375$), tier execution, nine-signal verification, the four-outcome decision under fixed cut-points $\tau_{store} = 0.60$ / $\tau_{defer} = 0.45$, and output; Stage 8 closes every cycle boundary with retroverify ($\tau_{retro} = 0.50$), deferred reconsider (TTL = 2), consolidate, LoRA fine-tune ($r = 32$, $\alpha = 64$, $\tau_{train} = 0.70$), and the multi-modal retention probe under floor $\rho_{min} = 0.93$ with asymmetric rollback (adapter reverted, memory preserved). Backbone Qwen-2.5-3B-Instruct, frozen across the cycle.

Per-Cycle Trajectory

Pooled headline Retention probes (ratio vs. pristine)						
Configuration	Cycle	n/bench	EM	CHM	MMLU TQA-test	CSQA-test
<i>Reference floor (no CAEM architecture; same Qwen-2.5-3B-Instruct backbone)</i>						
B1 (zero-shot)	—	300	0.507	0.120	—	—
<i>CAEM trajectory (full architecture: memory + verifier + SIL + retention guard)</i>						
CAEM	C0	300	0.541	0.135	—	—
CAEM	C1	300	0.543	0.112	1.041	1.013
CAEM	C2	300	0.544	0.107	0.975	<u>0.949</u>
CAEM	C3	300	0.532	0.115	0.992	0.987
CAEM	C4	300	0.532	0.115	1.018	0.886
CAEM	C5	300	0.517	0.109	0.999	<u>0.958</u>
Headline contrast (CAEM C2 vs. B1 zero-shot): $\Delta EM = +0.037$ (+7.4% relative); $\Delta CHM = -0.013$ (−10.8% relative).						

Headline Result

Against the matched zero-shot baseline B1 on the same backbone, the training-panel architectural-contribution slice at the within-trajectory best cycle (C2) reaches a pooled exact-match gain of +20.3% relative and a pooled composite hallucination metric reduction of −29.4% relative; the trajectory-end cycle holds at +13.7% EM and −20.3% CHM. CAEM beats every external baseline on both axes at both reference cycles.

$$\Delta EM = +20.3\% \text{ rel.} \quad \Delta CHM = -29.4\% \text{ rel.}$$

(training panel, CAEM C2 vs. B1 zero-shot baseline)

External Baseline Comparison

# System	EM	CHM	vs. CAEM C2 (best)		vs. CAEM C5 (final)	
			ΔEM	ΔCHM	ΔEM	ΔCHM
CAEM C2 (best)	0.544	0.107	—	—	+0.027	−0.002
CAEM C5 (final)	0.517	0.109	−0.027	+0.002	—	—
B1 zero-shot	0.507	0.120	+0.037 (+7.4%)	−0.013 (−11.0%)	+0.010 (+2.0%)	−0.011 (−9.3%)
B2 chain-of-thought	0.500	0.123	+0.044 (+8.8%)	−0.016 (−12.7%)	+0.017 (+3.4%)	−0.014 (−11.0%)
B3 RAG	0.458	0.148	+0.086 (+18.8%)	−0.041 (−27.7%)	+0.059 (+12.9%)	−0.039 (−26.4%)
B4 CoT + RAG	0.461	0.142	+0.083 (+17.9%)	−0.035 (−24.6%)	+0.056 (+12.1%)	−0.033 (−23.2%)
B5 five-shot CoT	0.405	0.136	+0.139 (+34.2%)	−0.029 (−21.4%)	+0.112 (+27.5%)	−0.027 (−19.9%)
B6 vanilla fine-tune (C5)	0.325	0.144	+0.219 (+67.2%)	−0.037 (−25.5%)	+0.192 (+58.9%)	−0.035 (−24.1%)
B7 FLARE	0.292	0.153	+0.252 (+86.3%)	−0.046 (−29.9%)	+0.225 (+77.1%)	−0.044 (−28.6%)
B8 semantic entropy	0.505	0.120	+0.039 (+7.7%)	−0.013 (−10.5%)	+0.012 (+2.4%)	−0.011 (−8.9%)

Wins-against count (CAEM beats baseline on ΔEM and ΔCHM jointly): CAEM C2 beats 8/8; CAEM C5 beats 8/8.

Retrieval-Utility Classifier Ablation

Benchmark	Exact match			Composite hallucination			Tier 3 share		
	Off	On	Δ (pp)	Off	On	Δ (% rel)	Off	On	Δ (pp)
<i>Training panel</i>									
FEVER	0.503	0.503	0.0	0.117	0.120	+2.5%	92.3%	70.7%	−21.7
TriviaQA	0.437	0.463	+2.7	0.123	0.121	−2.0%	95.7%	68.0%	−27.7
CommonsenseQA	0.603	0.743	+14.0	0.131	0.086	−34.4%	57.7%	20.3%	−37.3
<i>Transfer panel</i>									
StrategyQA	0.617	0.640	+2.3	0.119	0.128	+8.1%	100.0%	76.7%	−23.3
TruthfulQA	0.210	0.310	+10.0	0.116	0.119	+2.9%	99.0%	40.0%	−59.0

Pooled (all five)	0.474	0.532	+5.8	0.121	0.115	−5.2%	88.9%	55.1%	−33.8
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